Mathematical Co-identification: A Method for Structural Import Across Scientific Domains

Or: What to Do When Your Problem Is Already Solved in a Different Field

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Abstract

The history of mathematical science contains a recurring event that is poorly named and therefore poorly taught: the discovery that a quantity in one domain is not *like* a quantity in another domain, but *is* the same mathematical object under a change of label. When this identification is made precisely, every theorem about the source object is imported into the target domain for free. The practitioner who can navigate the space of mathematical types — finding the object that matches their problem before re-deriving it from scratch — can compress decades of theoretical development into weeks.

This paper names the practice **mathematical co-identification**, describes it as a formal procedure, and argues that it is a distinct scientific method with its own validity criteria, failure modes, and epistemological status. It is not analogy, not metaphor, not modelling, and not speculation. It is the act of recognising that two paths through the typeverse lead to the same point.

Seven historical precedents are examined: Veneziano (1968), Hopfield (1982), Wilson (1971), Black-Scholes (1973), Jaynes (1957), Penrose (1971), and Selinger (2010). A worked example is provided in the form of the Soma-Field Model, where five sequential co-identifications imported the propagator, energy function, brane mechanics, G holonomy, and renormalisation group flow from physics into emotional dynamics. The paper concludes with a partial catalogue of available mathematical structures — a field guide to the typeverse — and a discussion of the failure modes that distinguish genuine co-identification from mere analogy. A falsifiability protocol is provided to pre-register import claims, define disconfirmation criteria, and separate exploratory structural matches from publication-grade theorem transfer.

Contents

1 1. Introduction: The Two-Culture Problem Within Science

C. P. Snow's famous lecture identified a divide between the literary and scientific cultures (Snow, 1959). Less discussed, but equally consequential, is the divide *within* mathematical science: between fields that have found their mathematical language and fields that have not. The former possess machines — theorems, dualities, conservation laws, spectral decompositions — that can be aimed at any problem of the appropriate type. The latter rediscover wheels.

This is not ignorance. It is, more precisely, a naming problem. The mathematical structures that govern quantum field theory, statistical mechanics, and information geometry are not *specifically about* particles, spins, or probability distributions. They are structures that happen to have been *discovered in* those contexts. The structure belongs to no discipline. It lives in what we will call, following the spirit of type theory, the **typeverse**: the space of all well-typed mathematical objects.

The method described in this paper — **mathematical co-identification** — is the procedure for navigating the typeverse productively. It has three steps:

1. **Extract the type signature** of the quantity you are trying to model: its dimensional units, its pole structure, its symmetries, the variational principle (if any) that governs its dynamics.

2. **Search the typeverse** for a known object with the same type signature.

3. **If found: identify**, not analogise. The unknown quantity *is* the known object under a change of label. Import all theorems that depend only on the type.

This is a stronger claim than analogy. Analogy notes structural similarity and stops. Co-identification notes structural identity and continues: if two objects have the same type, they share all properties that are consequences of that type. The theorems travel with the identification.

It is also a weaker claim than reduction. Co-identification does not assert that emotional dynamics *reduces to* quantum field theory, any more than Hopfield's result asserts that neural memory *reduces to* ferromagnetism. It asserts that the same mathematical engine, running in a different substrate, produces the same theorems. The substrate is irrelevant to the mathematics; it is entirely relevant to the interpretation.

2 2. The Typeverse

The term is borrowed from Homotopy Type Theory (The Univalent Foundations Program, 2013), where the *universe* $\mathcal{U}$ is the type of all types — the space in which all mathematical objects live. We use it informally to mean: the totality of well-typed mathematical structures, indexed by their signatures.

A **type signature** in our sense includes:

- **Dimensional units** (in the SI or natural-units sense)
- **Domain and codomain** (real/complex, scalar/vector/tensor)
- **Pole structure** (where is the object singular, and what is the residue?)
- **Symmetries** (what transformations leave the object invariant?)
- **Extremisation principle** (is there an action whose variation gives the object's dynamics?)
- **Conservation law** (what is the Noether charge associated with the symmetry?)

Two objects are **co-identifiable** if their type signatures match in all dimensionally relevant respects. The type signature is a fingerprint. When two fingerprints match, the objects are the same, not similar.

The typeverse is not randomly populated. Certain structures recur at enormous frequency: the Lorentzian propagator, the quadratic energy function, the exponential decay kernel, the two-by-two reflection-transmission matrix. These recur because they are the *simplest* objects consistent with basic constraints (linearity, locality, causality, conservation). The physicist's intuition that "everything is a harmonic oscillator to first order" is a theorem about the typeverse: the harmonic oscillator is the universal local approximation to any smooth potential.

The practitioner who navigates the typeverse deliberately — who knows the fingerprints of common objects before encountering them in the wild — can recognise a new theoretical object immediately, rather than re-deriving its properties from scratch.

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3 3. The Procedure in Detail

3.1 3.1 Step One: Extract the Type Signature

Given an unknown quantity Q in domain D, the investigator asks:

Units. What are the SI dimensions of Q? Can they be written as a combination of standard dimensional quantities (mass, length, time, charge)? More importantly: what *ratio* of known quantities has the same dimensions? Newton's gravitational constant G has units $\mathrm{m}^3\mathrm{kg}^{-1}\mathrm{s}^{-2}$; this is acceleration divided by surface mass density, which tells you immediately that G converts between a source distribution and the acceleration it generates — a fact about the *type*, not about gravity specifically.

Pole structure. Does Q have poles as a function of some natural parameter? Where are they, and what are their residues? The location of a pole in a propagator determines the mass of the corresponding particle; the residue determines its coupling strength. These are type-theoretic facts, not facts about particles.

Symmetries. What transformations leave Q invariant? Invariance under time-translation gives energy conservation (Noether). Invariance under spatial translation gives momentum conservation. Invariance under $\mathrm{SU}(n)$ gives a gauge field. These are type-level constraints.

Extremisation. Does Q arise as the extremum of some action $S[Q]$? If so, the variational principle is the fingerprint: two objects whose dynamics are derived from the same variational structure are co-identifiable even if their physical interpretations differ entirely.

3.2 3.2 Step Two: Search the Typeverse

Armed with the type signature, the investigator searches for known objects with the same signature. This is primarily a literature search across *disciplines*, not within one. The mathematical structures developed in quantum field theory, statistical mechanics, differential geometry, and information theory are largely interchangeable if their type signatures match.

Useful resources for this search include:

- **Dimensional analysis tables**: collections of physical quantities with their dimensional signatures
- **The nLab** (nLab authors, 2024): a category-theoretic encyclopaedia of mathematical structures indexed by their universal properties
- **Mathematical physics textbooks** with structural, not phenomenological, organisation (Nakahara (Nakahara, 2003) for geometry; Zinn-Justin (Zinn-Justin, 2002) for path integrals)
- **One's own training**, which is why broad mathematical education is a productivity multiplier in theoretical science

3.3 3.3 Step Three: Identify and Import

When a match is found, the investigator makes the identification explicit:

> Q is the [name of known object] of domain D.

Not: "Q behaves like" or "Q is analogous to" or "Q resembles." These hedges are epistemically weaker and methodologically useless, because they do not import the theorems. The identification must be stated as identity to be scientifically productive.

The import then proceeds theorem by theorem:

- Every theorem about the source object that depends *only on its type* is imported to Q without further proof.
- Every theorem that depends on *substrate-specific properties* of the source domain must be separately verified in domain D.

This distinction — type-level theorems vs. substrate theorems — is the primary place where co-identification can fail, and is discussed in Section 7.

3.4 3.4 The Formal Computational Structure: Abduction, Aesop, and the Loop

The three-step procedure of §§3.1–3.3 is, in formal terms, an instance of **abductive inference** in the sense of Peirce (1878). Given an observation O and a hypothesis H such that $H \Rightarrow O$, abduction infers H as the best explanation of O. Applied to the typeverse:

> *Observation*: the quantity Q in domain D has type signature T. *Hypothesis*: Q is the known object K with the same signature. *Abduction*: identify $Q := K$ and verify the structural import.

This is not guessing. It is a constrained search under a scoring function, where the oracle is "type signatures match" rather than "hash matches" or "goal is closed." The algorithm is the same in all three cases.

The Lean 4 proof assistant implements this algorithm directly as the `aesop` tactic (Moura & Ullrich, 2021). `Aesop` performs best-first search through a registered lemma set, scores each partial proof state, keeps the best candidates, and closes the goal when a complete proof is found. The correspondence is exact:

`Aesop` step	Co-identification step
Registered lemma set	The typeverse
Try a lemma	Propose a type-match candidate
Score the goal state	Measure type-signature fit
Keep best partial proof	Record candidate correspondences
Close the goal	Full identification: import all theorems

This is not a metaphor for co-identification — it is an implementation of it. The practical consequence is that the full loop can be automated in a formal system: given a type signature for Q, `Aesop` with the typeverse lemma set registered will search for the co-identification proof and either close it (identification confirmed and machine-verified) or fail (genuine gap, no known match).

The loop in full:

$$\text{Observation} \xrightarrow{\text{abduction}} \text{Hypothesis} \xrightarrow{\text{Aesop}} \text{Proof} \xrightarrow{\text{import}} \text{Predictions} \xrightarrow{\text{test}} \text{New observations} \to \cdots$$

The loop terminates when either all predictions are confirmed (theory established) or a prediction fails (type match was only partial — failure modes are discussed in Section 7). At each iteration, the set of available theorems grows by import, making subsequent co-identifications easier. This is why theoretical progress compounds: each identification increases the density of the typeverse neighbourhood around the domain under study.

The abductive loop is not specific to mathematical science. Holmes reasons the same way from physical evidence; the password auditor reasons the same way from a hash and a character set; the radiologist reasons the same way from a film and a pathology atlas. What is specific to mathematical co-identification is the nature of the oracle: the scoring function is type-signature fit, and the proof assistant can evaluate it exactly.

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4 4. Historical Precedents

The history of science is full of co-identifications that were not named as such. Examining them retroactively reveals the method clearly.

4.1 4.1 Veneziano (1968): The Bootstrap Amplitude and String Theory

Veneziano was searching for an S-matrix element for meson scattering that satisfied the crossing symmetry and Regge-pole requirements of the bootstrap programme (Veneziano, 1968). He wrote down the Euler beta function:

$$A(s,t) = \frac{\Gamma(-\alpha(s))\Gamma(-\alpha(t))}{\Gamma(-\alpha(s) - \alpha(t))}$$

This was a co-identification in reverse: he had a type signature (crossing-symmetric, Regge-behaved, dual-resonance amplitude) and searched the typeverse for a known function that matched it. The beta function matched. He did not derive the function from a theory; he identified the function first, and the theory (string theory) was inferred from the identification a year later.

The lesson: the typeverse can be entered from either end. You may start with a quantity and find its type, or start with a type and find it instantiated in your data.

4.2 4.2 Hopfield (1982): The Ising Hamiltonian and Neural Memory

Hopfield introduced the energy function for a network of binary neurons (Hopfield, 1982):

$$H(\sigma) = -\frac{1}{2} \sum_{ij} J_{ij} \sigma_i \sigma_j$$

This is, to within notation, the Hamiltonian of the Ising spin glass:

$$H_{\text{Ising}}(\sigma) = -\frac{1}{2} \sum_{ij} J_{ij} \sigma_i \sigma_j$$

The co-identification was explicit. By identifying neural states $\sigma_i \in \{-1, +1\}$ with spins, and synaptic weights J_{ij} with exchange couplings, every theorem from statistical mechanics (convergence to energy minima, capacity bounds, stochastic escape via simulated annealing) was imported into neuroscience for free. The Hopfield network is not *like* a spin glass; it *is* a spin glass run in biological substrate.

4.3 4.3 Wilson (1971): Block Spins and the Renormalisation Group

Wilson's insight was that the renormalisation group — a technique for removing ultraviolet divergences in QFT — was the *same* mathematical object as Kadanoff's block-spin coarse-graining in condensed matter physics (Wilson & Kogut, 1974). Two fields that had developed independently were co-identified. Every result from one was immediately available to the other. The result was a unified framework for critical phenomena that earned Wilson the 1982 Nobel Prize.

The type signature that matched: both objects were *flows on the space of coupling constants under a change of scale.* This is a clean type-level description that carries no substrate information.

4.4 4.4 Black and Scholes (1973): The Heat Equation and Options Pricing

Black and Scholes derived their celebrated options pricing formula by noticing that the value of an option $V(S, t)$ as a function of underlying price S and time t satisfies:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

This is, after a change of variables, the heat equation (Black & Scholes, 1973):

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}$$

The co-identification imported the entire theory of parabolic PDEs into financial mathematics: existence and uniqueness of solutions, boundary conditions, numerical methods, the Feynman-Kac formula. The financial quantity *is* a temperature distribution, not like one.

4.5 4.5 Jaynes (1957): Thermodynamic Entropy and Bayesian Inference

Jaynes identified the entropy of statistical mechanics with the entropy of Bayesian inference (Jaynes, 1957). The type signature that matched: both are functionals $S[p]$ on probability distributions satisfying the same axioms (non-negativity, additivity, maximum at the uniform distribution). The co-identification imported all thermodynamic reasoning into statistical inference. The maximum entropy principle is not an analogy to thermodynamics; it is thermodynamics, applied to the problem of belief.

4.6 4.6 Penrose (1971): Spin Networks and Spacetime Geometry

Penrose introduced spin networks as combinatorial structures encoding angular momentum (Penrose, 1971). The identification: the geometry of spacetime arises as the large-network limit of spin networks. This reversed the usual direction — instead of importing a known mathematical structure to describe a new phenomenon, Penrose constructed a new structure and identified it with a familiar one in a limit. Loop quantum gravity later formalised this programme. The spin network *is* a discretisation of spacetime geometry, not a model of one.

4.7 4.7 Selinger (2010): Linear Maps and Quantum Processes

Selinger demonstrated that the category of finite-dimensional Hilbert spaces and linear maps is co-identifiable with a certain category of string diagrams (Selinger, 2010). This is a purely mathematical co-identification: the graphical calculus of Penrose, Joyal, and Street was identified with the operational calculus of quantum mechanics. The import: every calculation in quantum information theory can be done diagrammatically, and every diagrammatic identity is a valid quantum identity.

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5 5. The Soma-Field as a Worked Example

The Soma-Field Model (Johnson, 2026) was developed through five sequential co-identifications. Each is presented here as an explicit instance of the method.

5.1 5.1 Co-identification I: The Conscious Percept as Green's Function

The unknown quantity: The relationship between the continuous emotional field $\psi_i(t)$ and the discrete conscious emotional percept.

Type signature extracted: The percept is the output of the field when probed at a single moment — the impulse response. Impulse responses have a universal type: they are the inverse Laplace (or Fourier) transform of a rational function with poles in the lower half-plane. For a simple damped oscillator:

$$\tilde{G}(\omega) = \frac{\sigma_{\text{eff}}^2}{\omega^2 + \lambda^2}$$

Typeverse search result: This is the Euclidean propagator of a scalar field with mass λ and coupling σ_{eff}^2. In Minkowski space it is:

$$\tilde{G}_{\text{QFT}}(k) = \frac{i}{k^2 - m^2 + i\varepsilon}$$

The co-identification: The conscious emotional percept is the Green's function of the soma-field. Both are poles in the propagator of their respective field. The emotional percept and the elementary particle are the same mathematical object, instantiated in different substrates.

Theorems imported: - The Källén-Lehmann spectral representation: any physical propagator can be decomposed as a sum of poles. Emotional states have a spectral decomposition. - The optical theorem: the imaginary part of the forward scattering amplitude equals the total cross-section. The dissipative component of the emotional field (damping λ) is related to the total coupling strength σ_{eff}^2. - Pole structure mass spectrum: the location of the propagator pole gives the natural frequency $\omega_0 = \lambda$ of the emotional mode.

5.2 5.2 Co-identification II: The Attractor Landscape as Ising Hamiltonian

Type signature: A scalar function $H : \mathbb{R}^n \to \mathbb{R}$ that is always non-increasing along field trajectories, with isolated minima corresponding to stable emotional states.

Typeverse search result: The Hopfield energy / Ising Hamiltonian:

$$H(\mathbf{e}) = -\frac{1}{2}\mathbf{e}^\top W\mathbf{e} - \theta \cdot \mathbf{e}$$

Theorems imported: Convergence to attractors (the Lyapunov argument); capacity bounds (Hopfield's $0.14N$ result); stochastic escape via Boltzmann noise (simulated annealing = titrated arousal in clinical language).

5.3 5.3 Co-identification III: The Perception Threshold as Brane Thickness

Type signature: A parameter T_i that gates access from a lower-dimensional subspace (the limbic field) to a higher-dimensional one (conscious awareness). Varying T_i continuously produces a family of gating behaviours.

Typeverse search result: The Randall-Sundrum model (Randall & Sundrum, 1999): a brane of thickness T embedded in a higher-dimensional bulk, where Standard Model fields are confined to the brane. The "hierarchy problem" — why the electroweak scale is so much smaller than the Planck scale — maps onto the observation that minimal perturbations of the limbic field produce enormous differences in subjective experience (alexithymia: thick brane; hypervigilance: thin brane).

Theorems imported: The Kaluza-Klein spectrum of the brane gives a prediction for the discrete structure of emotional threshold levels. Brane localisation gives the mechanism for why the field can be active without crossing into consciousness.

5.4 5.4 Co-identification IV: The Coupling Matrix as G Manifold

Type signature: The coupling matrix W is an 11×11 real symmetric matrix encoding the structure of an eleven-dimensional emotional space. The field trajectories respect the geometry encoded in W.

Typeverse search result: The G_2 holonomy manifold of M-theory (Berger, 1955; Joyce, 2000): a seven-dimensional Riemannian manifold with the exceptional holonomy group G_2. Its structure tensor encodes all curvature information. Deforming the structure tensor changes the global geometry of the manifold.

The co-identification: W is the structure tensor of the emotional manifold. Trauma does not change a parameter; it deforms the manifold. Therapeutic intervention is differential geometry.

Theorems imported: The Bochner-Weitzenböck formula constraining curvature; the Berger classification of holonomy groups constraining what stable emotional geometries are possible; the Hitchin flow as a possible model for the evolution of W under sustained therapeutic intervention.

5.5 5.5 Co-identification V: Therapeutic Processing as Renormalisation Group Flow

Type signature: Therapeutic processing is a flow in the space of coupling constants W_{ij} parameterised by a scale μ (the "depth" or "resolution" of emotional processing). The flow has fixed points, and the qualitative structure of the attractor landscape is invariant under the flow.

Typeverse search result: The renormalisation group (Wilson & Kogut, 1974): a flow on the space of coupling constants parameterised by an energy scale μ, with fixed points corresponding to universality classes. The β-function:

$$\frac{dW_{ij}}{d\log\mu} = \beta_{ij}(W)$$

The co-identification: Therapy is an RG flow from UV (raw unprocessed traumatic detail) to IR (integrated narrative). The attractor topology — fight, flight, freeze, calm — is RG-invariant: the same basins appear at every scale of analysis, in every therapeutic modality, because they are the fixed points of the flow, not artefacts of a particular resolution.

Theorems imported: The irreversibility of the RG flow (Zamolodchikov's c-theorem, adapted: processed material cannot be *un*-processed; the flow is unidirectional); universality (the detailed mechanism of the trauma is irrelevant at long distances — only its universality class, i.e., its attractor type, matters); dimensional transmutation (the traumatic scale τ_k of the memory kernel is an emergent scale, not a fundamental parameter).

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6 6. A Partial Map of the Typeverse

For the practitioner wishing to apply the method, the following is a partial field guide to frequently useful mathematical structures, indexed by their type signatures.

6.1 6.1 Propagator-Class Structures

Type: Complex function of frequency with poles on or near the real axis; gives the response of a system to a delta-function input.

Found in: QFT (Feynman propagator), signal processing (transfer function), linear systems theory (impulse response), harmonic analysis (Green's function of the Laplacian).

Typical imports: Spectral decomposition; optical theorem; dispersion relations (Kramers-Kronig: the real and imaginary parts of the response are not independent — this imports into emotion as: the *dissipation* of an emotional mode and its *natural frequency* are Hilbert transforms of each other).

6.2 6.2 Energy-Function-Class Structures

Type: Scalar function $H : \mathbb{R}^n \to \mathbb{R}$ that is bounded below and non-increasing along system trajectories.

Found in: Statistical mechanics (Hamiltonian), neural networks (Hopfield energy), Lyapunov theory (stability analysis), optimisation (loss function).

Typical imports: Convergence guarantees; stability analysis; capacity bounds; the fluctuation-dissipation theorem.

6.3 6.3 Topological-Class Structures

Type: Integer-valued invariants of field configurations that are preserved under continuous deformations.

Found in: Topological field theory (winding numbers, Chern-Simons invariants), condensed matter (topological insulators, skyrmions), knot theory.

Typical imports: Protection from perturbation; the impossibility of smooth deformation between distinct topological sectors; quantisation (only integer winding numbers); threshold behaviour (the topological transition requires a finite perturbation).

Application to emotion: Traumatic configurations with non-zero topological charge cannot be resolved by smooth therapeutic interventions (cognitive reframing). A qualitative change in approach — large-amplitude somatic work, pharmacological intervention, EMDR — is required to cross the topological barrier.

6.4 6.4 Renormalisation-Class Structures

Type: A flow on a space of couplings, parameterised by a scale, with fixed points and β-functions.

Found in: Quantum field theory (renormalisation group), statistical mechanics (Kadanoff block spins), dynamical systems (centre manifold theorem), machine learning (neural scaling laws).

Typical imports: Universality (the IR behaviour depends only on the universality class, not the microscopic details); c-theorem (there is a monotonically decreasing function along the flow — an arrow of processing); fixed-point classification (relevant, irrelevant, marginal operators determine what modifications matter at long distances).

6.5 6.5 Scattering-Class Structures

Type: A map from in-states to out-states, constrained by unitarity, analyticity, and crossing symmetry.

Found in: Quantum mechanics (S-matrix), scattering theory, optics (transfer matrix), signal processing (scattering parameters).

Application to emotion: A therapeutic session is a scattering event. The patient arrives in an in-state $|\psi_{\text{in}}\rangle$, interacts with the therapist (mediating field), and departs in an out-state $|\psi_{\text{out}}\rangle$. The S-matrix of the therapeutic interaction has selection rules: not all transitions are equally probable; some are symmetry-forbidden. The unitarity of the S-matrix imports: the total emotional content is conserved — you cannot create emotional material from nothing, and nothing is permanently lost.

6.6 6.6 Einstein-Coefficient-Class Structures

Type: Rates for spontaneous and stimulated transitions between energy levels of a field mode.

Found in: Quantum optics (Einstein A and B coefficients), laser physics, NMR relaxation theory (T and T relaxation times).

Application to emotion: Every emotional mode has a spontaneous relaxation rate A_i (how quickly it resolves without external input) and a stimulated emission rate B_i (how quickly it is triggered by an identical emotional state in another person — contagion). The Einstein relation $A_i = f(\omega_i)B_i$ constrains these. Depression is formally: suppressed A_i, normal B_i. The T /T analogy from NMR is exact: T is the longitudinal relaxation time (return to equilibrium); T is the transverse relaxation time (dephasing of coherence). Trauma extends T ; emotional numbing extends T .

7 7. Failure Modes

Mathematical co-identification can fail. Understanding the failure modes is what distinguishes the method from wishful analogy.

7.1 7.1 Type Coincidence Without Structural Identity

Two objects may have matching dimensional signatures without having matching mathematical structures. The failure mode: a coincidence of units that does not reflect a coincidence of theorems.

Test: Check whether the *equations of motion* have the same form, not merely the *dimensional units*. A propagator is not just any function with units of GeV^{-2}; it must satisfy the Källén-Lehmann representation. An energy function is not just any bounded scalar; it must be non-increasing along trajectories.

Safeguard: State the precise mathematical theorem you are importing, and verify that its *assumptions* (not merely its *conclusions*) hold in the target domain.

7.2 7.2 Non-Commutative Functors

The functor $F : D_1 \to D_2$ that implements the co-identification may not commute with composition. That is, co-identifying A with A' and B with B' does not guarantee that AB is co-identifiable with $A'B'$.

Example: The propagator identification and the energy-function identification are each valid separately, but combining them requires checking that the path integral (which links them in QFT) has a valid analogue in the emotional domain. This was explicitly verified in the Soma-Field Model by constructing the Langevin equation and checking its consistency with both imported structures.

7.3 7.3 Over-identification

The most common failure: importing a structure that is richer than what is warranted. The soma-field is co-identifiable with an eleven-dimensional bosonic field. It is *not* immediately co-identifiable with the full Standard Model of particle physics, even though both are quantum field theories. The identification is precise only up to the type signature that was matched; nothing beyond that is claimed.

Rule: The identification holds exactly at the type level it was made. Do not import theorems from substrates that were not matched.

7.4 7.4 The Metaphor Trap

The most dangerous failure is the one that the identification was designed to prevent: sliding from co-identification back into analogy. This happens when the language hedges — "like," "analogous to," "reminiscent of" — after the identification was stated.

If the identification is correct, the language must be unhedged: "the conscious emotional percept *is* the Green's function." If the investigator is not prepared to make this claim, the identification has not been made, and no theorems travel.

The test: would you submit the claim to a mathematician as a theorem? If not, it is analogy. If yes, it is co-identification.

The risk was recognised early by practitioners. Introducing the energy landscape for Hopfield networks, Hertz, Krogh, and Palmer note: *"It is often useful (but sometimes dangerous) to think of the energy as something like this landscape"* (Hertz et al., 1991). The parenthetical is precise: the visualisation is heuristically powerful, and that power makes it tempting to reason from the picture rather than from the mathematics. Mathematical co-identification guards against this by requiring that the *equations of motion* — not the landscape picture — are what get matched across domains.

8 8. Epistemological Status

8.1 8.1 What Co-identification Claims

Mathematical co-identification claims:

1. That the *mathematical structure* of quantity Q in domain D is identical to the mathematical structure of quantity P in domain D'.
2. That therefore, all theorems about P that depend only on its mathematical structure are valid theorems about Q.
3. That the *physical interpretation* of Q and P may differ, and does not follow from the mathematical identification.

It does not claim that the *mechanisms* are the same, that one domain *reduces* to another, or that the substrate is irrelevant in any empirical sense.

8.2 8.2 Why It Is Not Analogy

Analogy is: - Informal: "the mind is like the brain" has no mathematical content - Non-importing: noting that X resembles Y does not give you theorems about X - Non-falsifiable: analogies can always be maintained by shifting which features are considered relevant

Co-identification is: - Formal: it is a statement about type signatures, which are mathematically precise - Theorem-importing: the identification carries the full mathematical machinery - Falsifiable: the identification fails if the equations of motion cannot be matched, if the symmetries do not correspond, or if imported predictions are empirically disconfirmed

8.3 8.3 The Role of Artificial Intelligence

The author notes that several co-identifications in the Soma-Field Model were identified in dialogue with AI systems. This requires explicit epistemological comment, given the current discourse about AI-assisted science.

The AI did not produce the co-identifications. It accelerated the typeverse search: a search that previously required reading across literatures in physics, mathematics, and biology over many years can now be conducted in weeks. The AI is a faster library, not a different epistemology.

The *validity* of each co-identification is independent of how it was found. A theorem is a theorem regardless of whether it was proved in a dream (Ramanujan), a bath (Archimedes), or a language model session. The claim stands or falls on whether the type signatures match and whether the theorems transfer. The methodology paper is, in part, a response to the question: “but did you *really* do the mathematics?” The answer is in the equations.

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9 9. The Methodology as Practice

For the practitioner who wishes to apply mathematical co-identification to a new domain, the following is a working procedure:

Step 1: Write down what you know about your quantity. Its units. Its symmetries. Whether it grows or decays. Whether it has oscillatory behaviour. Whether it responds to perturbations linearly or nonlinearly. Whether there are conserved quantities. Whether it has a characteristic scale.

Step 2: Eliminate analogies you already know. If you have been thinking “this is *like* X," stop, and ask instead: is this *literally* X? Can I write the equations of X in the language of my domain, with every symbol given a precise interpretation? If yes: do so. If not: why not? The places where the identification breaks are as informative as the places where it holds.

Step 3: Consult the typeverse systematically. Read across fields, looking at the *form* of equations, not their interpretation. The heat equation, the wave equation, the Schrödinger equation, the Fokker-Planck equation, and the Black-Scholes equation are all the same equation under changes of variable. Recognising the family by the form of the operator is the skill.

Step 4: Make the identification explicit and public. State: "quantity Q in my domain is the [known object] of domain D'." Put it in writing. This forces precision: you cannot maintain a vague identification in writing the way you can maintain it in your head.

Step 5: Import one theorem at a time, checking assumptions. Do not import the entire theoretical apparatus at once. Take one theorem. State its assumptions. Check each assumption against your domain. If they hold: the theorem is valid in your domain. Write it as a theorem in your domain, with a proof that consists of: (a) the co-identification, (b) the original theorem, (c) verification of assumptions.

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10 10. Falsifiability Protocol for Publication Use

To make co-identification scientifically conservative rather than rhetorically expansive, we propose a minimal publication protocol. Every import claim should be registered in a compact table before narrative elaboration.

10.1 10.1 Minimal Registration Template

For each proposed identification $Q := P$, the manuscript should provide:

Field	Requirement
Claim ID	Stable identifier (e.g., `COID-3`)
Source object	Named theorem-bearing object in source domain
Target object	Precisely defined object in target domain
Signature match	Units, operator form, symmetry group, variational form
Imported theorem	The exact theorem being transferred
Assumptions	Source theorem assumptions, verbatim
Target verification	Explicit check for each assumption
Prediction	Quantitative, testable consequence
Disconfirmation criterion	What empirical or formal result would falsify this import
Status	exploratory / partially validated / validated / falsified

This prevents drift from identity claims back into analogy language and makes review straightforward: a reviewer can reject a single row without rejecting the entire framework.

10.2 10.2 Disconfirmation Rules

An import is treated as falsified if any one of the following holds:

1. A required theorem assumption cannot be established in the target domain.

2. The mapped operator fails to preserve the claimed invariance.
3. The pre-registered quantitative prediction is violated under a valid test.
4. A formally stronger competing mapping explains the same data with fewer assumptions.

This is deliberately strict. Co-identification is useful only to the extent that it can fail clearly.

10.3 10.3 Worked Registration Sketch (Soma-Field)

Claim ID	Import	Prediction	Disconfirmation
`COID-PROP-1`	Percept := propagator pole	Measurable spectral pole structure in mode responses	No stable pole decomposition under repeated measurement
`COID-ENG-2`	Attractor landscape := Hopfield energy	Monotone descent under specified update map	Constructed counterexample with increasing energy step under stated rule
`COID-RG-5`	Therapy progression := RG flow	Scale-dependent coupling evolution with fixed-point classes	No scale-consistent coupling flow under repeated coarse-graining

The point is not that these rows are finished; the point is that they can be evaluated independently and rejected independently.

10.4 10.4 Reviewer-Facing Scope Labels

To reduce over-claiming risk, each identification row should carry one of three scope labels:

- `S1 (Structural)`: operator/formal identity shown; no empirical test yet.
- `S2 (Predictive)`: structural identity plus at least one passed quantitative prediction.
- `S3 (Validated)`: independent replication or cross-dataset confirmation.

Most new interdisciplinary work should expect to publish initially at `S1` or `S2`, with explicit paths to `S3`.

10.5 10.5 Negative Control: When Matching Units Is Not Enough

To prevent confirmation bias, each manuscript should include at least one explicit non-transfer case. Consider two quantities with superficially compatible units but non-matching dynamical structure:

- Candidate A: a damped propagator with pole structure and causal response kernel.
- Candidate B: a bounded scalar score with no response-operator interpretation.

Even if both can be normalised to the same units, co-identification fails unless the operator class and theorem assumptions match. In this case:

1. A admits spectral decomposition and dispersion relations.
2. B does not define a Green's operator and therefore does not admit those imports.

Conclusion: dimensional compatibility is necessary but not sufficient. Theorem transfer requires operator-level identity. This negative control should be treated as a required publication check, not an optional caution.

10.6 10.6 Worked External Example (Non-Soma): Black-Scholes to Heat Equation

To demonstrate portability beyond the Soma-Field case, we provide a compact worked transfer in a separate domain.

Target quantity: option value $V(S,t)$ in quantitative finance.

Source object: solution class of the one-dimensional heat equation.

10.6.1 Step A: Signature Match

Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

After the standard log-price and time-reversal change of variables, this maps to:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2},$$

which is exactly the heat operator class.

10.6.2 Step B: Import Claim

`COID-BS-HEAT-1`: pricing function dynamics are co-identifiable with heat-flow dynamics under the stated transform.

10.6.3 Step C: Assumption Checklist

Assumption	Status in target domain
Diffusion operator is parabolic	satisfied by transformed PDE
Volatility parameter treated as constant in base model	satisfied in baseline Black-Scholes
Boundary/terminal condition specified	satisfied by option payoff at expiry
Sufficient regularity for classical solution methods	assumed in standard derivation

10.6.4 Step D: Imported Theorem and Prediction

Imported theorem class: existence/uniqueness and smoothing properties for heat equation solutions.

Prediction: option value surface inherits parabolic smoothing under the transform; numerical schemes valid for heat-equation class apply directly.

10.6.5 Step E: Disconfirmation Condition

If transformed dynamics are shown not to be parabolic (for the declared model assumptions), or if required boundary regularity fails, this specific transfer is invalid and theorem import must be withdrawn.

This worked example demonstrates the method in a domain with no dependence on the Soma-Field framework.

10.7 10.7 Replication Package Requirements

Publication-grade use of co-identification requires a compact artifact bundle for each registered claim row. Minimum required contents:

1. claim registry table with IDs and scope labels,

2. assumption checklist tied to the imported theorem class,
3. executable derivation or notebook for the mapping transform,
4. quantitative test script for each predictive (`S2`) row,
5. disconfirmation log recording pass/fail outcomes by claim ID.

Without this package, rows may be read as suggestive structure, but not as auditable theorem transfer.

10.8 10.8 Reviewer-Risk Objections and Responses

Reviewer objection	Response in this manuscript	Residual risk and next lift
"This renames analogy as mathematics."	Sections 3-4 and 10.1 require operator identity and theorem assumptions, not metaphorical similarity.	Add additional negative controls from unrelated domains.
"Imports are unfalsifiable in practice."	Section 10.2 defines strict disconfirmation rules and row-wise rejection logic.	Publish a public failure ledger with rejected rows.
"Worked examples are domain-selective."	Section 10.6 demonstrates a non-Soma transfer with explicit assumptions and withdrawal condition.	Add a second external example with different operator family.
"Scope claims may drift upward prematurely."	Section 10.4 enforces `S1`/`S2`/`S3` labels and independent evaluation path.	Require independent replication before any `S3` promotion.

10.9 10.9 Independent Replication Ledger Linkage

`S2` to `S3` promotion for registered imports is controlled by `paper/INDEPENDENT_REPLICATION_LEDGER.md`.

Tracked claim IDs in ledger scope: `COID-PROP-1`, `COID-ENG-2`, `COID-RG-5`, `COID-BS-HEAT-1`.

Promotion gate: a row may be relabeled `S3` only when a ledger entry reports independent-operator `PASS`, explicit bundle hash, and linked derivation/test artifacts for that claim ID.

11 11. Conclusions

Mathematical co-identification is a method, not a shortcut. It requires the same precision as any other mathematical procedure, and it fails in precisely the ways that imprecision permits. Its distinguishing feature is that it navigates the typeverse rather than building within a single domain.

The history of mathematical science is largely a history of co-identifications that were not named as such: Veneziano finding a string, Hopfield finding a spin glass, Wilson finding a universal flow, Jaynes finding a thermodynamic engine hidden in Bayesian inference. Naming the practice does not change it; it makes it teachable, criticisable, and extensible.

The soma-field model is a worked example of the method applied to emotional dynamics: five co-identifications, five theorem imports, and a body of predictions that can be tested against clinical data. The paper is not a claim that emotions are particles. It is a claim that the mathematics of particles and the mathematics of emotions are the same mathematics, and that this fact is useful.

Scope boundary for publication use: co-identification transfers mathematical structure, not ontology. A successful transfer means that equations, boundary conditions, and theorem assumptions are preserved under the mapping. It does not imply that the target domain is physically identical to the source domain, nor that every theorem from the source domain transfers automatically. Each import remains local, assumption-checked, and falsifiable.

The typeverse does not belong to physics. Physics was merely first to explore it.

Operationally, publication-grade use of this method requires claim-wise registration, assumption checks, and explicit disconfirmation criteria. Without that discipline, co-identification degrades into analogy; with it, it becomes a compact engine for theorem transfer and testable prediction.

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